

FATHIMA SHARSOOM

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Summary

Data Analyst with 3+ years of experience, possessing a strong foundation in statistics, machine learning, and programming. Proven ability to transform complex data into actionable insights, supporting strategic decision-making and data integrity.

Technical Skills

Languages: Python, SQL, Java, R-programming

Libraries: Pandas, NumPy, Scikit-Learn, Seaborn, Matplotlib, Pytorch, PySpark, TensorFlow

Frameworks and Tools: PowerBI, Tableau, AWS, PostgreSQL, MySQL, RStudio, Spark, Snowflake

Technologies: EDA, ML, Visualization, NLP, BigData, DL, ETL, Hadoop, DataMining, PredictiveAnalytics, Statistics

Certifications

- [IBM Machine Learning with Python, March 2024](#)
- [AWS Certified Cloud Practitioner, September 2024](#)

Professional Experience

Graduate Research Assistant, University at Buffalo

Aug 2024 – Current

- Analyzed 50+ eye-tracking datasets to evaluate the correlation between visual attention and surgical performance
- Created visual dashboards in Tableau highlighting trends based on experience, clinical background, and procedure types.
- Applied sliding window techniques for feature extraction to extract features such as fixation, saccade, entropy, and NNI
- Built ML models to predict practitioners' observational skill to identify critical surgical steps based on gaze patterns

Java Developer, Tata Consultancy & Systems

Mar 2021 - Aug 2023

- Developed and maintained architecture for REST API in Java for an eCommerce enterprise application
- Spearheaded the redesign and regrouping of APIs, resulting in a 20% increase in overall user satisfaction
- Utilized MySQL to manage, and resolve database issues, ensuring system performance and data integrity
- Improved application performance 30 % by optimizing source code and caching operations using RabbitMQ
- Collaborated with other teams to analyze system solutions and directly engaged with client to capture requirements
- Integrated Amazon S3 and AWS Lambda streamlining data handling and boosting platform efficiency
- Supported Business Analysts in translating business requirements into user stories, comprehensive technical documents

Full Stack Java Developer Intern, I Prime (via Marlabs)

Dec 2020 - Feb 2021

- Leveraged RESTful web services and Angular 8 for seamless front-end integration, and utilized JDBC to efficiently connect and manage MySQL databases
- Developed extensive expertise in a diverse array of back-end and front-end technologies
- Executed hands-on web development tasks, demonstrating proficiency in coding and debugging

Projects

Advancing Cervical Cancer Diagnosis: Deep Learning-Based Pap Smear Analysis

May 2024

- Designed and implemented ResNet50-Transformer model for classifying cervical cells, achieving accuracy of 92.85%.
- Implemented model pruning to optimize the ResNet50-Transformer model for faster inference.

Comparative Analysis for Customer Support Ticket Prioritization | *R, Predictive modeling, NLP*

Nov 2023

- Developed and implemented a predictive model to enhance the efficiency of customer support processes.
- Assessed machine learning models (Decision Tree, Random Forest, SVM, XGBoost, KNN), addressing dataset bias, enhancing performance, notably achieving 92% accuracy for XGBoost.
- Utilized visualization techniques to enhance interpretability and insights throughout the project.

E-Commerce Customer Segmentation | *Python, RFM analysis, Business Intelligence*

Nov 2023

- Implemented K-means clustering and RFM analysis, uncovering six distinct customer segments
- Achieved an accuracy of 80.27% by selecting six clusters through the Elbow, Silhouette, and Gap Statistic methods
- Applied the Pareto Principle to identify and prioritize the top 20% of revenue-generating customers, contributing to tailored marketing strategies for improved engagement and satisfaction.

Space Efficient Automated Packaging & Arrangement | *Python, Open CV*

Mar 2020

- Devised and constructed an automated mechanism, utilizing image processing to sort objects based on dimensions
- Utilized Open CV and Python to extract information from QR codes, to determine the destination of packages

Education

Master of science, Industrial Engineering Focus in Data Analytics, University at Buffalo, GPA 3.96

Dec 2024

Bachelors in Instrumentation and Electronics (Honors), APJ Abdul Kalam University, GPA 3.5

Oct 2020